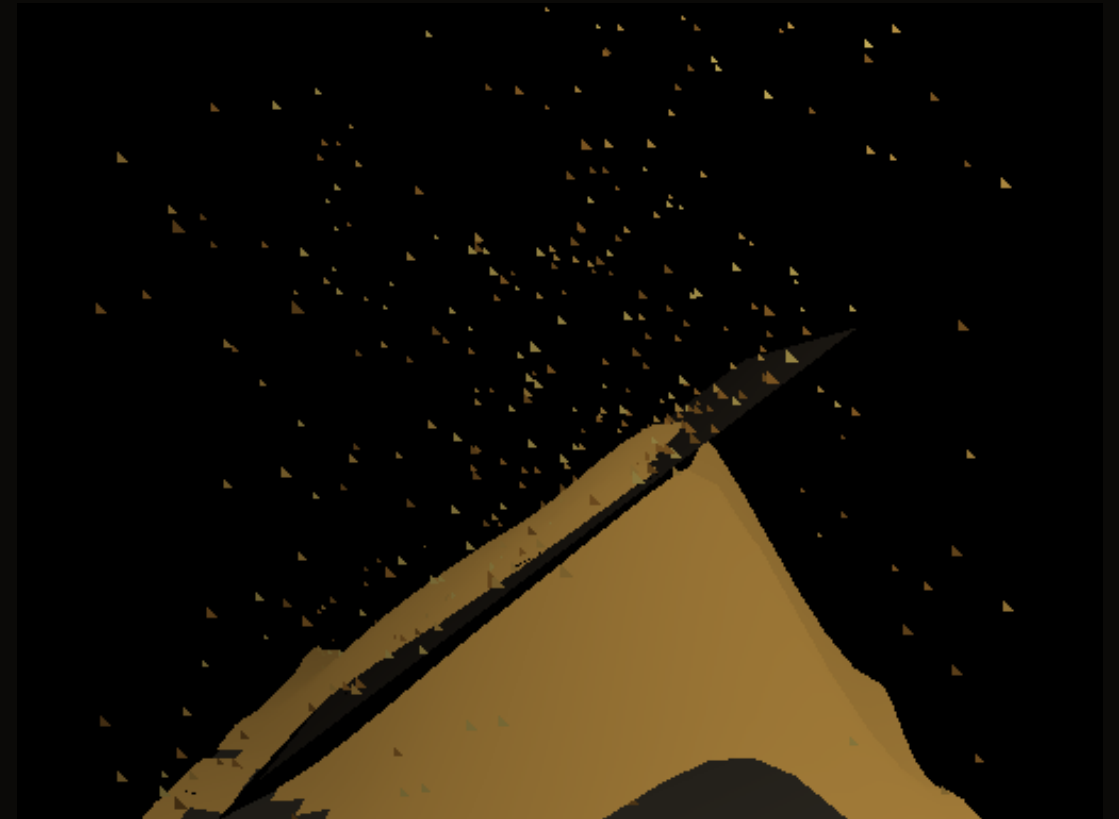


Grid-Accelerated Sand Pile Simulation

CS488 Final Project Demo

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A simplified particle-based sand model with spatial grid collision, ramp interaction, settling, height-field reconstruction, and angle-of-repose relaxation.



Final scene: sand particles, ramp obstacle, reconstructed pile surface

Demo roadmap: show objectives in proposal order

Target video length: about 3–5 minutes. No recompiling or editing during the demo.

1 Emitter + particle state

sand source, gravity, radius, active/settled state

2 Spatial grid + local collisions

grid cells, neighbor checks, debug coloring

3 Ground + ramp collision

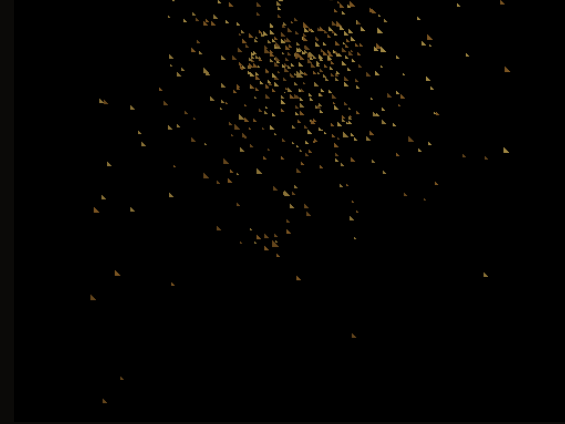
friction, low restitution, sliding obstacle

4 Settling + surface reconstruction

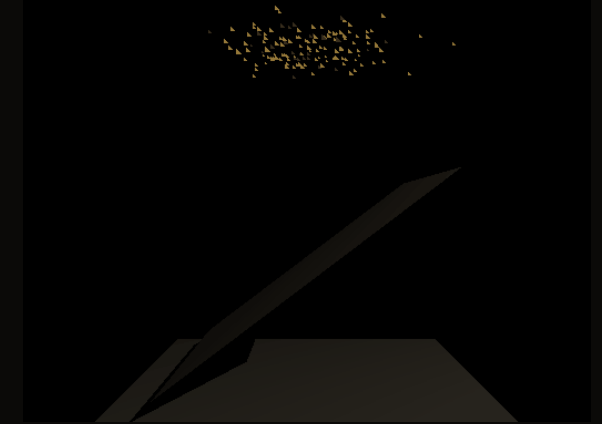
stable pile + height-field mesh

5 Angle-of-repose + limitations

slope relaxation, honest model limits



Grid debug view



Final surface view



Ramp collision view

System pipeline

The final project extends the A3 particle system into a sand-specific simulation pipeline.

Emitter → Verlet integration → Grid insertion → Local collisions → Settling → Height-field surface → Rendering



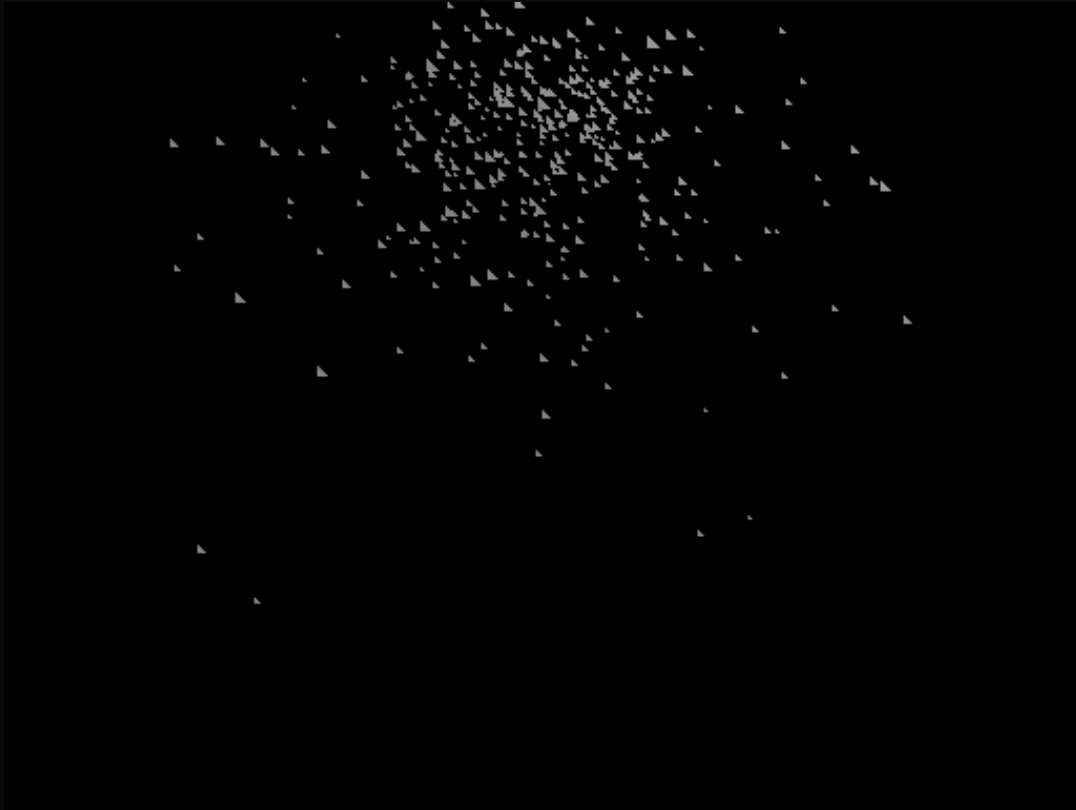
Live simulation result

What changes over time

- Particles spawn from a rectangular sand source.
- The spatial grid keeps particle interaction local.
- Ground and ramp collisions use position correction and friction.
- Settled particles become the input for the reconstructed sand surface.
- Angle-of-repose relaxation stabilizes steep height-field slopes.

Objectives 1–2: sand emitter and particle state

Goal: create sand particles with enough state to support falling, collision, and settling.



Milestone 1: sand source and falling particles

Implemented

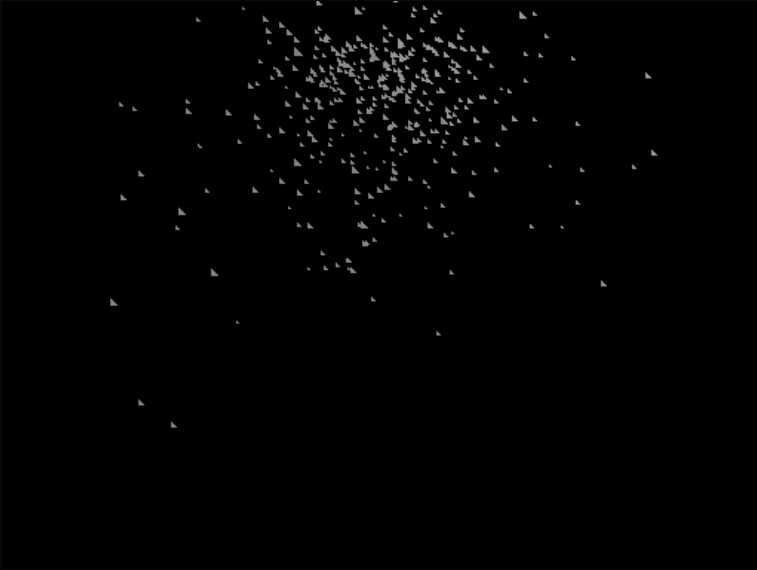
- Rectangular emitter near the top of the scene.
- Randomized position, initial velocity, radius, and age.
- Verlet-style update using current and previous positions.
- Each particle tracks active state and settled state.
- Low restitution and damping support sand-like behavior.

Code references

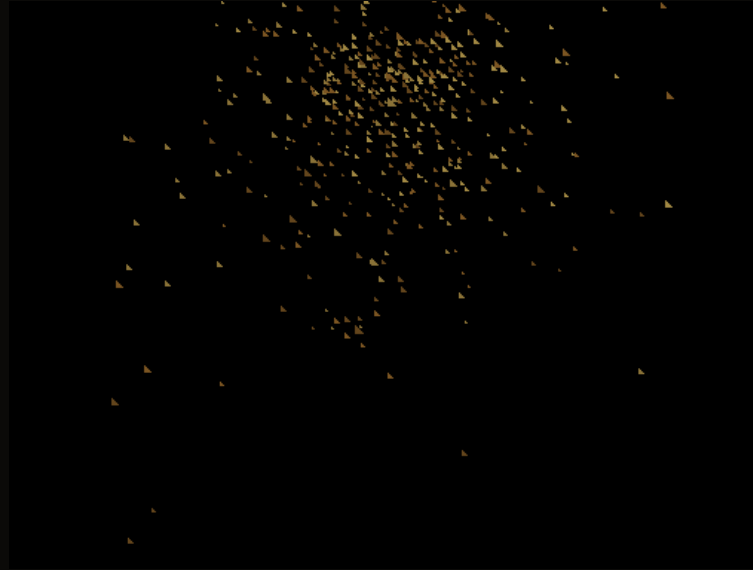
```
Particle::emitFromSandSource  
Particle::integrate  
ParticleSystem::emitParticles
```

Objectives 3–4: spatial grid and local particle collisions

Goal: avoid global all-pairs checks and resolve only local particle interactions.



Grid collision implementation



Grid debug coloring

Analytical optimization data

500

particles in final setup

124,750

all-pairs candidates

18^3

grid cells

27

neighbor cells/query

Collision response

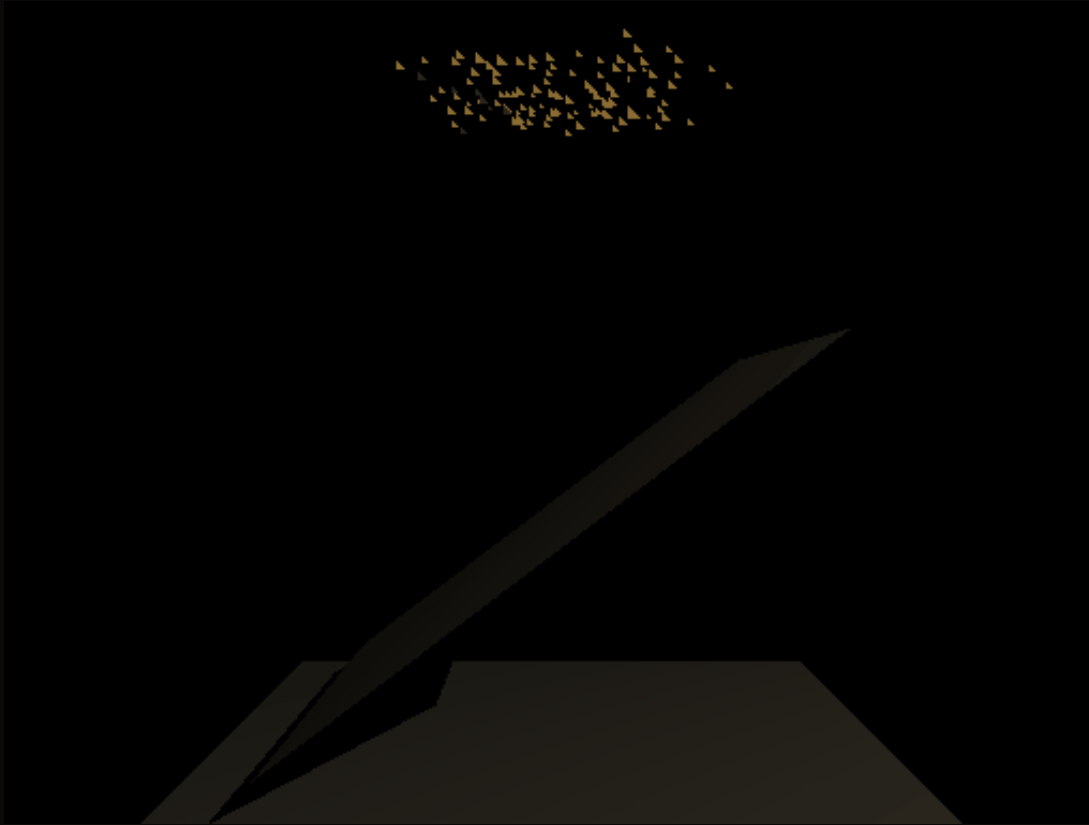
- Particles are treated as small spheres.
- Overlap is resolved by separation along the center-to-center normal.
- Velocity damping reduces unstable bouncing and interpenetration.

Implementation

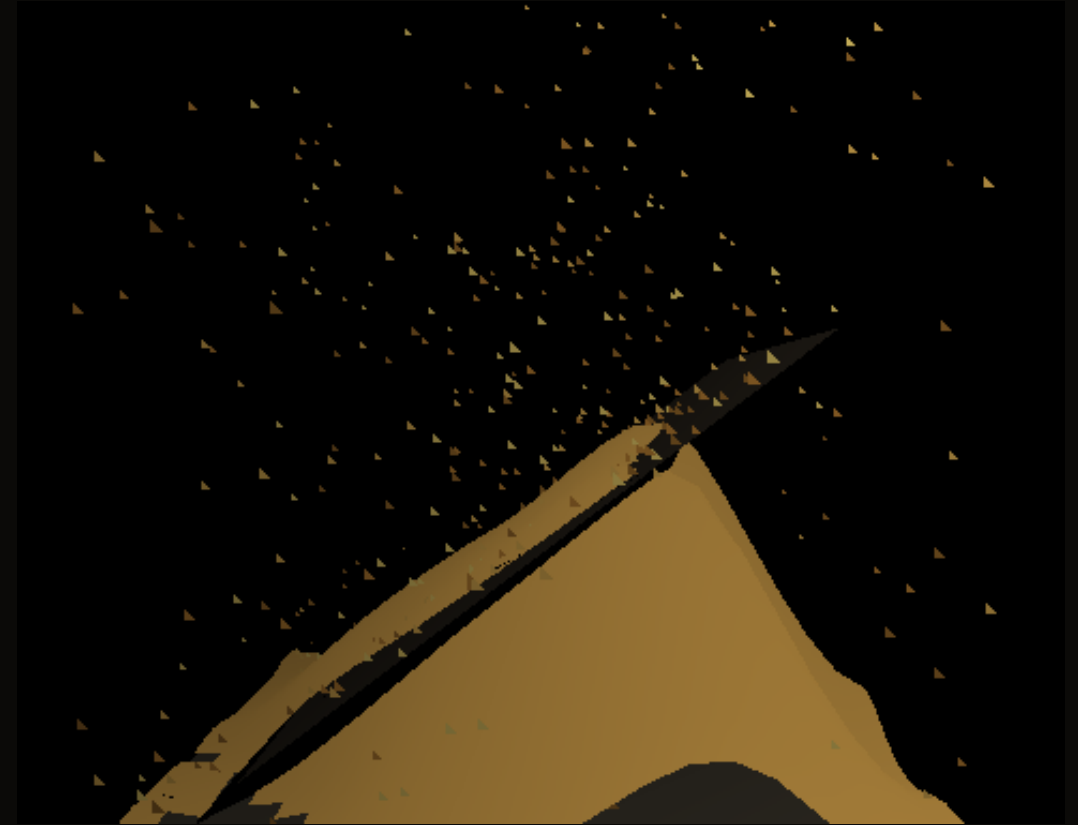
```
buildSpatialGrid()  
resolveSandParticleCollisions()  
resolveSandParticlePair()
```

Objectives 5–6: ground collision, friction, and ramp obstacle

Goal: demonstrate sand interacting with static geometry instead of falling through the scene.



Milestone 3: visible ramp and ground



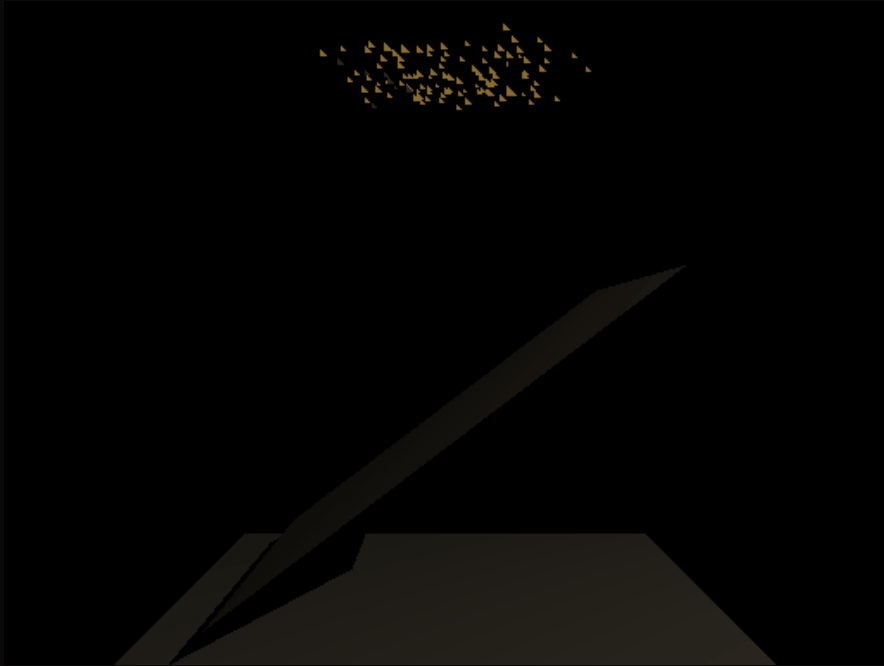
Final scene after ramp interaction

Collision behavior to point out

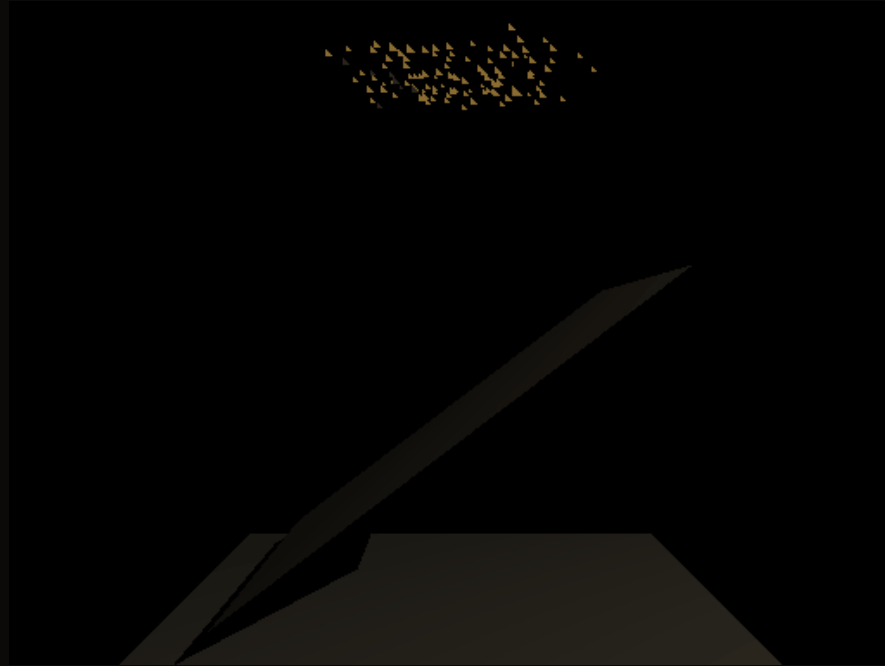
position correction • low restitution • tangential friction • ramp sliding

Objective 7: settling creates a stable pile

Goal: reduce jitter and let low-speed supported particles become part of the pile.



Before: moving particles dominate



After: settled particles contribute to pile

Settling rule

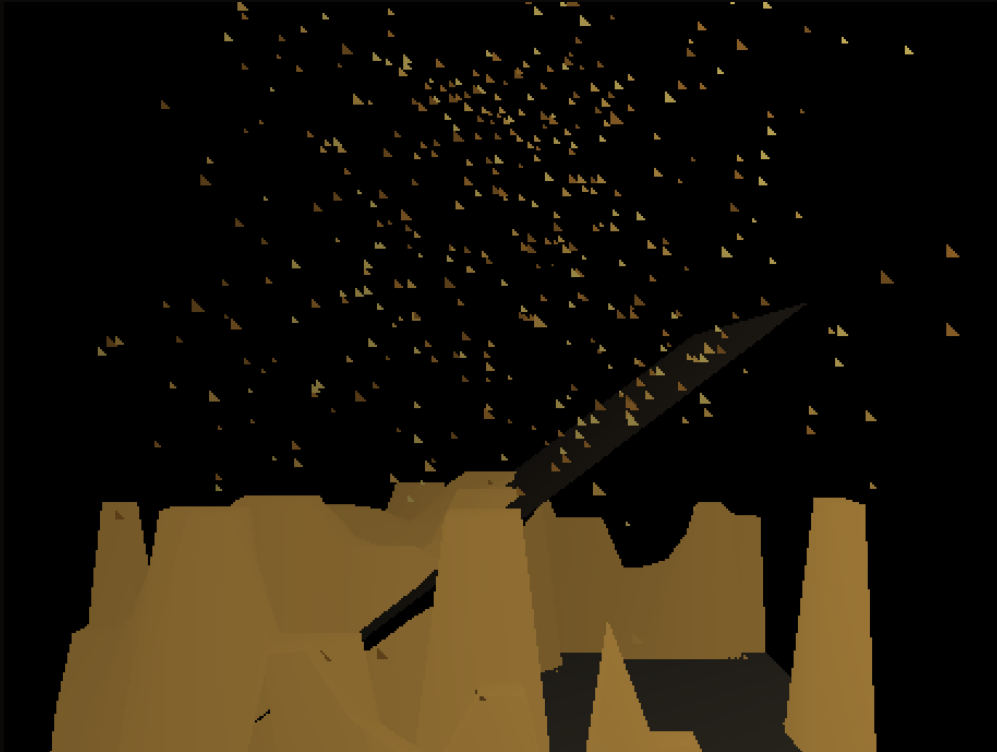
- old enough
- low speed
- supported by floor, ramp, or settled particle
- then velocity becomes zero

Why it matters

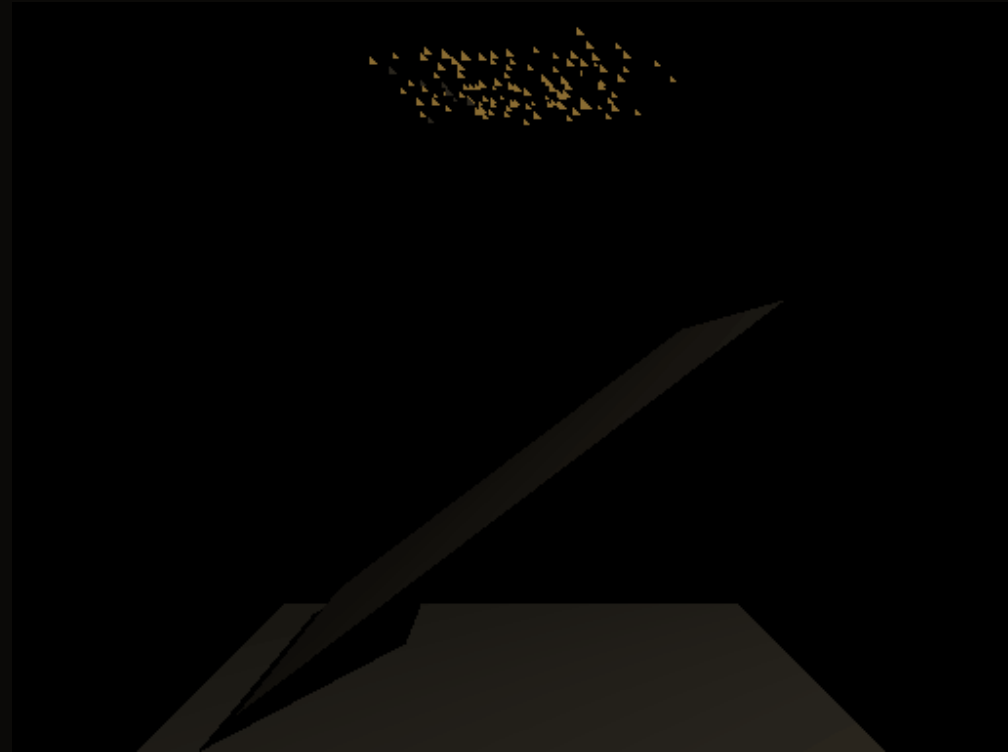
Without settling, particles jitter and the reconstructed surface is unstable.

Objective 8: height-field surface reconstruction

Goal: turn settled particles into a continuous visible sand-pile surface.



Earlier rough surface: blocky columns



Improved surface: settled-only + smoothed

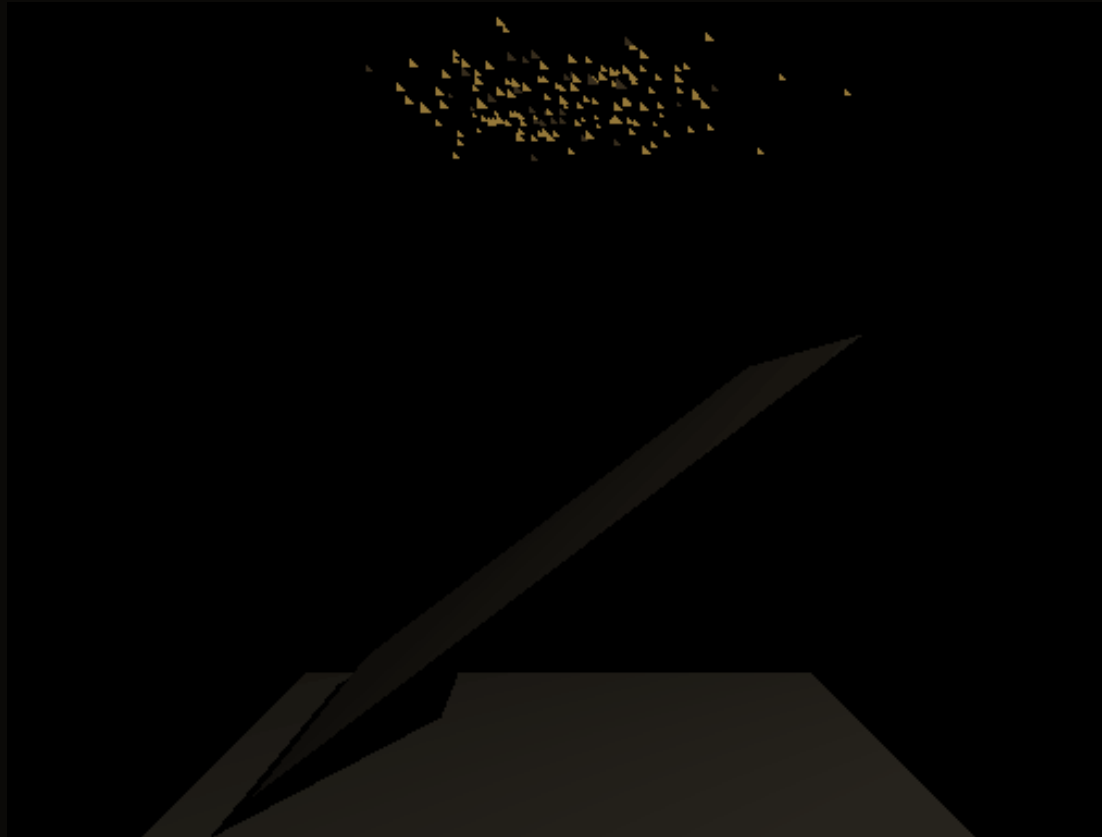
Surface pipeline

settled particles → x-z height field → smoothing → triangles

Uses heightFieldResolution = 24

Extra feature: angle-of-repose relaxation

Goal: give the reconstructed sand pile a simple physics-inspired maximum-slope behavior.



Milestone 5: relaxed surface slope

Simplified sand physics idea

- Sand piles usually stabilize below a maximum slope.
- I use 34° as the angle-of-repose parameter.
- If neighboring height cells differ too much, height transfers from the higher cell to the lower cell.
- This relaxation is applied to the reconstructed height field, not directly to every particle.

Rule used

$\text{max height difference} = \tan(34^\circ) \times \text{cell size}$

Video recording checklist

Use this slide while recording the demo video.

Before recording

- Do not recompile or edit code during the video.
- Open Terminal directly in `final_project`.
- Run `./build/CS488` once and show the live simulation.
- Then show the prepared images/GIFs for each objective.
- Keep the video around 3–5 minutes.

Suggested exact order

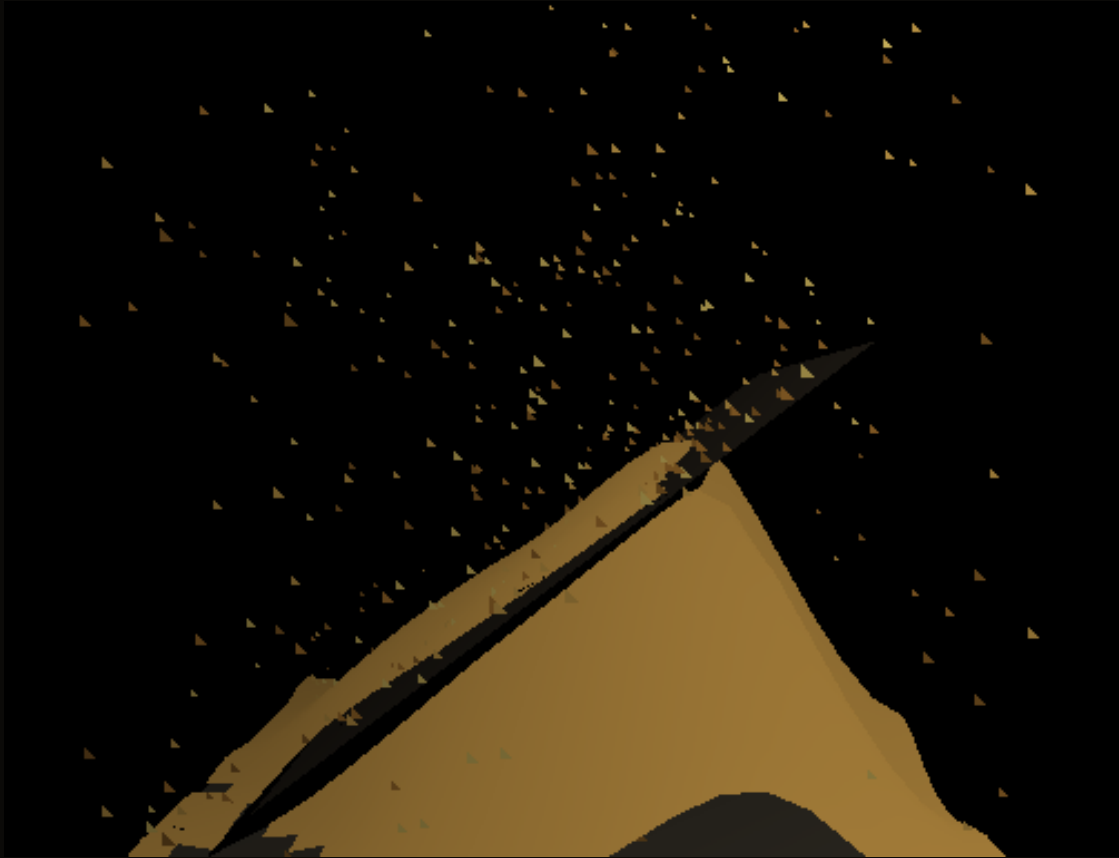
- Live run: final scene appears.
- `milestone1_sand_fall.gif`
- `milestone2b_grid_debug.gif`
- `milestone3_ramp_obstacle.gif`
- `milestone4_surface_reconstruction.gif`
- `milestone5_angle_of_repose.gif`
- Close with limitations.



Final files: README.md, screenshot.png, video.mp4, source code

Closing: what the project demonstrates

Strong points and honest limitations.



Final output

Strong points

- Spatial grid makes collision queries local.
- Particles collide with ground, side boundaries, and a visible ramp.
- Settling turns moving particles into stable pile material.
- Height-field reconstruction gives a continuous surface.
- Angle-of-repose relaxation adds a sand-specific slope model.

Limitations

This is a simplified particle + height-field model, not a full continuum granular solver. The angle-of-repose rule affects the reconstructed surface rather than the particles directly.